

TrES-2b

R-2509

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_110812 · created 2026-08-25T00:05:45+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2455785.889263 +/- 9.9e-05 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.135 +/- 0.043
Transit depth (Rp/Rs)^2	1.83 +/- 1.17 [%]
Orbital Inclination (inc)	84.49 +/- 2.83
Airmass coefficient 1 (a1)	2.02 +/- 1.33
Airmass coefficient 2 (a2)	-0.53 +/- 0.21
Scatter in the residuals of the lightcurve fit is	55.67 %
Best Comparison Star	#1 - [130, 296]
Optimal Aperture	2.44
Optimal Annulus	9.26
Transit Duration (day)	0.071 +/- 0.039

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.135 $\pm$ 0.043 vs 0.1254 (deviation 0.22 σ)
Duration fit vs published	0.071 d vs 0.0746 d (deviation 0.09 σ)
Mid-time O–C	-0.2 min
Depth SNR	3.1
Residual scatter	55.67 %

Light curve

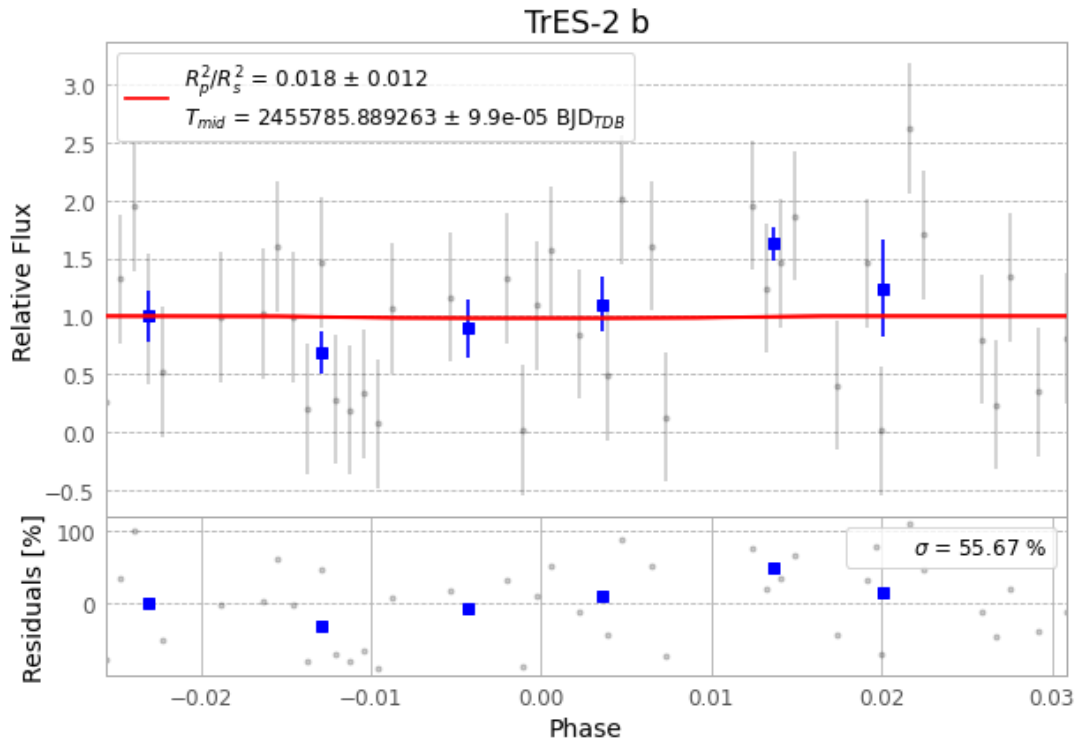


Figure 1 — FinalLightCurve_TrES-2 b_2011-08-12.png (SHA-256 7b01c4277e29324f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [300, 405], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 92abce0e98ba5d9f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

68 FITS frames (45 MB), TRES-2110812074512.FITS → TRES-2110812110612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-110812.log (f6a5b7535f97...), FinalLightCurve_TrES-2 b_2011-08-12.png (7b01c4277e29...), FinalLightCurve_TrES-2 b_2011-08-12.csv (4ecaaf8f8df1...)

Compute resources

Wall time 143.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-25 00:05Z.